

Water Displacement in Shafts and Geysering Created by Uncontrolled Air Pocket Releases

Kathleen Z. Muller¹; Jue Wang²; and Jose G. Vasconcelos, M.ASCE³

Abstract: Uncontrolled air pocket release from water-filled shafts can lead to geysering in stormwater systems. Such occurrences are deleterious from public health and environmental standpoints and can cause property and structural damage. Causes, frequency, magnitude, and location of geysering events still are poorly understood and pose practical difficulties to designers regarding how to create dropshafts that are less likely to present this issue. This paper presents results from experimental and numerical investigations on air-related geysers that provide insight into the mechanisms of air release and the displacement of water in vertical shafts. A 302-mm schedule 40 clear PVC apparatus is constructed with the essential features of a stormwater tunnel, and is fitted with vertical shafts with diameters ranging from 0.10 to 0.20 m. Predetermined air pocket volumes are released in the horizontal pipe and eventually reach shafts, causing water displacement and often causing geysers. Kinematics of the air pocket release are measured along with pressures at selected points in the apparatus. These results are used in the calibration of a computational fluid dynamics (CFD) model, which compares well with the experimental measurements. The model subsequently is used in a larger geometry that allows the evaluation of air pocket release kinematics for a wider range of conditions. Findings of this work provide further details on the nature and strength of geysering events, and suggestions for future studies in this topic are also provided. DOI: [10.1061/\(ASCE\)HY.1943-7900.0001362](https://doi.org/10.1061/(ASCE)HY.1943-7900.0001362). © 2017 American Society of Civil Engineers.

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Introduction

Stormwater systems can present a variety of operational problems when subjected to rapid inflow conditions associated with intense rain events. Such operational problems include excessive surging, overflows, large pressures, and manhole lid displacements, among others. In such conditions, these systems can create negative impacts on the environment and to public health, as well as costs associated with related structural damages. Large, entrapped air pockets in stormwater systems have the potential to worsen some of these problems. Air initially is present in stormwater systems prior to the filling process, and is displaced by water inflows and released through available ventilation points. Based on experimental observations when inflow rates are gradual, pressurization interfaces are near horizontal and air pocket formation is not observed (Patrick and Vasconcelos 2015). However, when rapid inflows are observed, various air pocket formation mechanisms can take place (Li and McCorquodale 1999; Vasconcelos and Wright 2006).

Once entrapped, air pockets move within tunnel systems as gravity currents (Benjamin 1968; Wilkinson 1982; Baines 1991), and this motion is subjected to effects such as background flows, drag forces, and so on (Chosie et al. 2014). When an air pocket of

significant volume arrives at the base of a water-filled ventilation shaft, these pockets will move vertically (Nielsen and Davis 2009; Vasconcelos and Wright 2011; Lewis 2011). This vertical air pocket motion creates the displacement of water ahead of the pocket, which in turn can lead to discharges of a mix of air and water through the tops of these shafts. This uncontrolled air pocket release phenomenon is referred to as a *stormwater geyser*, often characterized by jets of air–water mix rising through manholes and other vertical structures. Such occurrences can be very damaging and result in a number of deleterious impacts. Despite contributions from previous studies on geysers, the conditions associated with more severe geysers still are not fully understood. Designers of stormwater systems do not have enough guidelines to conceive dropshaft designs that are safe from the impacts of uncontrolled air pocket release.

Guo and Song (1991) presented an initial investigation of the phenomenon of geysers, and were among the first to observe that intense rain events led to the rapid filling of the Tunnel and Reservoir Plan (TARP) system in Chicago, which in turn created geysers. The mechanism they presented did not consider the possibility of two-phase flows, so geysering would be a result of mass oscillation processes within tunnel reaches. Although a feasible mechanism, Wright et al. (2011) pointed out that this mechanism would not be consistent with some observed geysering episodes, which are characterized by large discharges of air. Wright et al. (2011) also presented field measurements of an actual geysering episode recorded in Minnesota. Although geysers were reported, piezometric pressure heads did not exceed grade elevation. This is an important and troublesome characteristic of geysering phenomena, because it indicates that even if surge levels do not exceed grade elevation, geysers may occur.

The rise of gas pockets in vertical liquid columns has been studied in other contexts to some extent, but not extensively in the context of unsteady stormwater flows. For instance, Batchelor (1967) presented a classical problem of the rise of a Taylor bubble, and pointed that experiments indicated that the velocity scale increased

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with vertical shaft diameter D , and was close to $0.34\sqrt{gD}$. This estimate, however, was consistently smaller than the rising air pocket velocity observations by Vasconcelos and Wright (2011) and Lewis and Wright (2012). The cause of this is unclear, but may be linked to unsteadiness of air pockets leaving horizontal pipes into vertical pipe, as well the process of air volume increase as it moves in the vertical shaft and decompresses. Guedes de Carvalho et al. (2000) presented another interesting experimental study related to the release of air phases in vertical pipes was presented by. They described flood instability as a phenomenon created when liquid film flow moving downward in the vertical pipes has its interface roughed by a central upward gas flow. As this countercurrent flow increases, the gas–liquid interface may show waves and liquid detachment from the downward flow, which travel upward with the gas flow. The shearing process created by flooding instability may be linked to the spray characteristics of stormwater geysers.

Other contributions to the understanding of geysering processes have involved numerical modeling of these episodes. The work by Guo and Song (1991) was possibly the first that attempted to describe the geysering phenomenon, albeit in single-phase discharge conditions. Alternative approaches were proposed that considered the case of the water displacement caused by the rise of an air slug. A model based on lumped inertia analysis was presented in Vasconcelos and Wright (2011) and further developed by Lewis and Wright (2012) for a case in which the air release occurs amid sustained water pressure in the system. The models were successful in representing the experimental conditions presented by the authors, but failed in attempting to describe the severity of actual geysering episodes.

Shao (2013) developed a two-dimensional (2D) computational fluid dynamics (CFD) model that used the volume of fluid (VOF) method to describe air release in shafts and geyser occurrence. It was determined that small pockets do not create large displacements, whereas larger air pockets create large displacements of water and significant air phase flows upon pocket breakthrough. Choi et al. (2014) presented another study that implemented CFD to study geysers, which applied results by Vasconcelos and Wright (2011) to describe the characteristics of geysers created by the release of air pockets in laboratory conditions. The model was able to qualitatively represent some of the measured flow features during the experimental geysering episodes. However, large-scale conditions of actual geysering events pose other modeling challenges (e.g., air-phase compressibility) that were not explored in either of these CFD investigations. An interesting study by Catano-Lopera et al. (2014) involved the use of CFD models to simulate operational issues with the TARP system in Chicago. However, they made no attempt to systematically evaluate flow conditions related to uncontrolled air releases for geometry parameters, initial water levels, and varying entrapped air pocket releases.

In summary, geysers are a relevant operational issue that may impact the operation of stormwater systems, but still are not fully understood. Experiments have been performed to evaluate these conditions and some of these investigations have provided relevant insights into this issue. There are difficulties in scaling these experiments considering both air and water phases, so large-scale experiments should be able to offer more details on this phenomenon, as well support the development of models. Model studies of geysering events have been limited and have not been compared with larger-scale experimental data nor field measurements.

Objectives

This paper presents results of experimental and numerical investigations on the geysering created by the release of air pockets of

finite volume from stormwater systems in pressurized flow conditions. The work has three main objectives: (1) to perform observations of these events in large-scale experiments on air pocket release, along with data collection; (2) to apply and calibrate a CFD model to describe the displacement of water in the vertical shaft during the experimental tests; and (3) to use the same CFD modeling strategy in larger-scale conditions to understand, among other characteristics, the general trend of water displacement created by air pocket release from these shafts. The methodology for the investigation is presented next, followed by a section presenting and discussing results of this research, and finally conclusions and recommendations for future work.

Methodology

Experimental Methods

The experimental apparatus was a scale model of a portion of a stormwater storage tunnel, shown in Fig. 1. The apparatus included a 13.7-m-long clear PVC pipeline, with diameter D_t equal to 302 mm. This near-horizontal pipeline had two reservoirs (volume 1.1 m^3) supported by wooden structures connected to both the upstream and downstream ends, which were used to sustain pressures in the system during the air pocket releases. Approximately 10.7 m from the upstream point of the horizontal pipeline there was a Tee connection, in which 4.9-m-long vertical shafts of varying diameters ($D = 102, 153, \text{ and } 202 \text{ mm}$) were installed. Most of the experiments involved water in quiescent conditions, with a few cases performed with background flows.

Attached near the upstream end of the 302-mm horizontal apparatus was a 6-m-long, 202-mm-diameter clear PVC spur pipeline that served as an air chamber to store a known volume of air. This air phase was set at preselected pressure values that represented the pressures in the water side of the apparatus prior to the air pocket release. Halfway within this air chamber, a knife gate valve allowed for volume of air V_{air} of either 100 or 200 L to be stored prior to release. Finally, a 302-mm knife gate valve was placed at the upstream end of the horizontal pipeline and was kept partially closed during experiments to prevent the released air pocket from escaping through the upstream water reservoir.

The following instruments were used in the experiments:

- Three piezo-resistive pressure transducers: Transducer P1 (Endevco 8510C-50, Meggitt, Irvine, California), rated at 0.34 MPa, installed 0.6 m from the upstream end of the horizontal pipeline, near the pipe invert; Transducer P2 (Endevco 8510B-15), rated at 0.10 MPa, installed 9 m from the upstream end of the horizontal pipeline, near the pipe invert; and Transducer P3 (Endevco 8510B-15), rated at 0.10 MPa, installed at elevations ranging from 1.5 to 1.7 m above the horizontal pipe centerline, flush with the vertical shaft wall.
- An NI-USB 6210 data acquisition board (National Instruments, Austin, Texas) sampling pressures at a frequency of 200 Hz;
- Two pressure manometers (Extech HD750, Waltham, Massachusetts), one immediately downstream from the air chamber and the other at the base of the vertical shaft;
- A Vectrino acoustic doppler velocimeter (ADV) (Nortek, Rud, Norway) sampling at 25 Hz, located 7.5 m downstream from the upstream end of the horizontal pipeline; and
- A digital HD camcorder, recording the advance of the air–water interface at 30 frames per second.

Initially the apparatus was filled to an intermediate level within the vertical shaft. A known air volume in the air chamber was pressurized to a level similar to that observed in the vertical shaft.

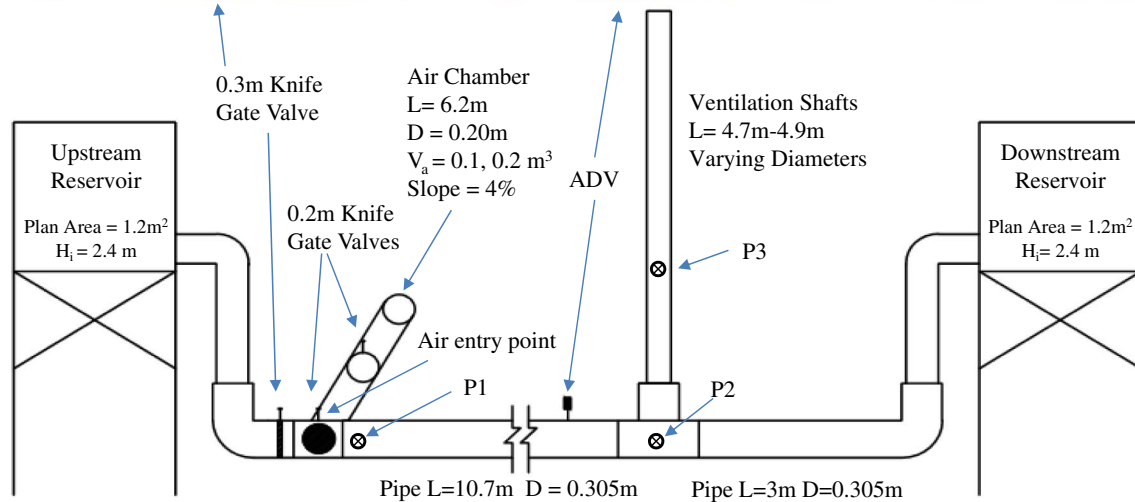


Fig. 1. Photo and sketch of experimental apparatus used in this investigation (image by Kathleen Muller)

Upon the opening of the knife-gate valve linking the air chamber to the water-filled horizontal pipe, the air pocket was released and advanced in the horizontal pipeline until it reached the base of the vertical shaft. Water levels in the shaft varied during the rise of the air pocket, as described by Vasconcelos and Wright (2011), and in various cases the level reached the upper end of the shaft, resulting in a geyser event. It was also observed, particularly for the smallest vertical shaft, that some of the released air continued to advance in the horizontal pipe while most of the air was released through the shaft.

The experimental procedure was performed as follows:

1. With the air chamber knife gate valve closed, the apparatus was filled with water to a piezometric head approximately 2.7 m with respect to the horizontal pipe inlet;
2. Water was in quiescent conditions prior to air release. In a few cases, background flows were created with recirculation pumps in the horizontal pipe;
3. The air compressor was turned on and the air pressure in the chamber was adjusted to approximately match the water pressure within the horizontal pipe in an attempt to reduce the amplitude of inertial oscillations in the ventilation shaft upon opening the air chamber valve;
4. Data collection from the pressure transducers, ADV, and video camera was started and the initial manometer readings were recorded;
5. The gate valve to the air chamber was quickly opened (<0.5 s) with one swift motion to allow the sudden release of the air pocket;
6. Upon release, the entrapped air pocket advanced in the system toward the vertical shaft, similarly to a cavity described by Baines (1991) and Chosie et al. (2014);

7. After approximately 9 s, the air pocket arrived at the base of the vertical shaft, causing the displacement of the water present in the shaft; and
8. After a steady state was reached, final pressure measurements were recorded with the manometers and all data recording devices were turned off.

Once the experimental runs were complete, the data analysis process was initiated. Videos were downloaded and visually analyzed to determine the vertical displacement of both the air pocket (air–water interface, or Y_{Int}) and the free surface of the water within the vertical shaft (referred to as Y_{FS}). This was done by progressing frame by frame through the videos and noting the time at which each surface reached the marks on the vertical shaft at 0.30 m intervals. These data then were used to calculate the velocity associated with Y_{Int} and Y_{FS} motion by dividing the distance traveled by elapsed times.

Values for vertical displacements and velocities, air pocket volume, and vertical shaft diameters were normalized, and henceforth in this paper starred variables (*) refer to the normalized versions of the variables. The vertical shaft diameter D was normalized by the horizontal pipe diameter D_r , whereas air pocket volumes V were normalized by D_r^3 . Vertical displacement measurements were normalized by the length of the vertical shaft L . Velocities of the free surface V_{FS} or the vertical air–water interfaces V_{Int} were normalized by $\sqrt{gD}$.

The proposed experiments used two main experimental variables: the ventilation shaft diameters and the volume of air pockets released. Three diameter values were used: $D^* = 0.33, 0.50$, and 0.67 (ranging from 102 to 202 mm). Two air pocket volumes were used: $V^* = 3.5$ and 7.0 (100 and 200 L). Some experiments were performed with background flows that had velocities measured

with the ADV sensor. Such velocities, in the range of 0.25 m/s, did not create significantly different results when compared with cases in which there were no background flows. This possibly is because background velocities were much smaller than those of large discrete air pockets, $0.54\sqrt{gD_r}$, which in the case of these experiments exceeded 0.9 m/s. Thus such results were omitted from this work. Most conditions were repeated at least three times in order to ensure consistency of the experimental results.

Computational Methods

Although the experimental tests were performed on a scale larger than most previous related studies, the apparatus dimensions were not comparable with those of actual stormwater systems. Because one overarching goal of the investigation was to evaluate the displacement of water created by large air pockets in stormwater systems, this work applied computational fluid dynamics modeling to help in the characterization of uncontrolled air pocket releases in water-filled shafts. Previous investigations (Choi et al. 2014; Catano-Lopera et al. 2014) have applied CFD to describe geysering processes. The present investigation, unlike previous contributions, used CFD to perform a systematic evaluation of air pocket releases for a wider range of geometries, water levels, and air pocket volumes. In order to achieve this, experimental results were used to support the development of an *OpenFOAM* CFD model, helping to identify more adequate formulations and mesh characteristics.

OpenFOAM, first developed by Weller et al. (1998) as a field operation and manipulation (FOAM) tool, is a free C++ library used to create applications for CFD. The flow solvers in *OpenFOAM* can be applied to both single-phase and multiphase flows with either laminar or turbulent Navier–Stokes equations. To conduct a two-phase flow simulation such as geyser events, *interFOAM* is one alternative solver that applies the VOF phase-fraction method (Hirt and Nichols 1981) to track air–water interfaces.

One limitation of *interFOAM*, however, is that it does not account for the compressibility of the air phase. As a result, early attempts in this investigation of simulating the process of air release and geyser events with *interFOAM* were not entirely successful. It was noticed that air compressibility was relevant even in the experimental scale used in the present experiments. An alternative *interFOAM* solver, *compressibleInterFOAM*, which is able to simulate nonisothermal immiscible flows, was adopted. The two-phase Navier–Stokes equations with VOF method solved by *compressibleInterFOAM* can be expressed as (Svenungsson 2016)

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{U}) = 0 \quad (1)$$

$$\frac{\partial (\rho \mathbf{U})}{\partial t} + \nabla \cdot (\rho \mathbf{U} \mathbf{U}) - \nabla \cdot (\mu \nabla \mathbf{U}) - \rho g = -\nabla p - F_s \quad (2)$$

$$\frac{\partial (\rho C_p T)}{\partial t} + \nabla \cdot (\rho \mathbf{U} C_p T) - \nabla \cdot (k \nabla T) = S_T \quad (3)$$

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \mathbf{U}) + \nabla \cdot [\alpha(1 - \alpha) U_r] = 0 \quad (4)$$

where ρ = average density of the fluid within a cell; $\mathbf{U}$ = flow velocity vector; μ = dynamic viscosity; g = gravity acceleration; p = pressure; F_s = surface tension force; C_p = specific heat; T = temperature; k = thermal conductivity; S_T = energy source term; α = fraction of water in the cell (between 1 and 0); and U_r = velocity field suitable to compress the interface. The average density is defined as

$$\rho = \alpha \rho_w + (1 - \alpha) \rho_a \quad (5)$$

with ρ_w = density of water; and ρ_a = density of air. Eqs. (1)–(3) represent mass, momentum, and energy conservations, respectively. Eq. (4) is related to the compression effects on the interface. A standard $k - \epsilon$ model was applied to resolve the turbulence, but the details of turbulence modeling are not presented here for brevity.

The CFD model in this paper was created with the main objective of representing the kinematics of the air–water release in the experimental conditions. The mesh was generated with *snappyHexMesh* (Greenshields 2015), which is a polyhedral mesh generator that chisels existing mesh into the required mesh. Three mesh resolutions were considered in the mesh independence study performed to represent the experimental conditions, using a 153-mm-diameter vertical shaft ($D^* = 0.50$). The approximate mesh cell counts in the 153-mm-diameter shaft were 5,000 (1.4 million in the whole mesh), 44,000 (2.2 million whole mesh), and 352,000 (3.9 million whole mesh) for coarse, intermediate, and fine mesh sizes, respectively. As the mesh was refined, numerical results converged to experimental results. Reasonably good agreement was achieved with the intermediate and fine mesh options, and intermediate mesh sizes were selected to perform most of the numerical simulations of this study.

Once adequate CFD model setup was achieved, particularly once discretization was determined, a series of hypothetical larger-scale simulations using larger tunnel geometry with varied sizes was performed using the same modeling strategy applied in the model representing experimental results. The hypothetical larger-scale geometry in general was similar to the geometry used in the experiments, with means to sustain the water pressure with a reservoir and the presence of a single ventilation shaft in the horizontal pipe. Two key geometric differences were the absence of an upstream reservoir and the much longer vertical shafts of 150 m and of varying diameters. The horizontal tunnel diameter was 1 m and the length in most simulations was set to 60 m, excluding the additional length that stored the initial air pocket volume. As is explained in the section “CFD Modeling Results,” in a few cases the horizontal tunnel length was increased.

Compared with the experimental tests, a wider range of conditions was simulated using the CFD model. Four different air pocket volumes were simulated, with three initial water depths in the vertical shaft and three different vertical shaft diameters. A total of 30 different conditions were tested (Table 1). For all simulations, it was assumed that the roughness of the tunnel and vertical shaft walls was smooth. It also was assumed that the reservoir and vertical shaft were open to the atmosphere and that the air and water temperature was 27°C.

Results and Discussion

Experimental Results

The tests began with an initial water level of approximately 2.7 m in the vertical shaft, measured from the invert of the horizontal pipe. The opening of the 202-mm knife gate valve released the air pocket. Even though the air pressure matched the initial water piezometric pressure in the apparatus, a few inertial oscillations were observed in the shaft upon the valve opening, which lasted a few seconds. These amplitudes were smaller than the vertical shaft diameter by the time the pocket, traveling at approximately 0.93 m/s, arrived near the base of vertical shaft.

In most of the tested cases the water displacement exceeded 2.5 m, which led to geysering through the top of the shaft. It is

Table 1. Range of Numerical Variables Considered in the CFD Simulations of Uncontrolled Air Release in Larger-Scale Geometries

Variable	Range considered	Normalized range
Initial air pocket volume	$V = 1.0, 5.0, 10, \text{ and } 50 \text{ m}^3$	$V^* = V/D_t^3 = 1, 5, 10, \text{ and } 50^a$
Initial water level at vertical shaft	$Y_{FS,0} = 10, 20, \text{ and } 50 \text{ m}$	$(Y_{FS,0}/L) = 0.067, 0.133, \text{ and } 0.333$
Diameter of the ventilation shaft	$D = 0.25, 0.50, \text{ and } 1.0 \text{ m}$	$D^* = D/D_t = 0.25, 0.50, \text{ and } 1.0$

^aLimited number of cases simulated with air pocket volume $V^* = 50$.

important to state that the geysering criterion used in this study was based on the release of a slug of water ahead of the rising air pocket rather than on the splash of water created once the air pocket reached the free surface. However, the releases were characterized by a significant mix of air and water, many times explosive in nature. In a few of the tested cases there was more than one discharge of water through the top of the shaft. Following the air–water discharge, a complex flow pattern with smaller slugs and bubbles rising was observed in the vertical shaft and remained there for a brief period (several seconds) along with observed inertial oscillations.

In all tested cases the displacement of Y_{FS} reached or exceeded 2.0 m. The corresponding normalized free surface displacement $\Delta Y_{FS}/D$ ranged from a minimum of 10 (for $D^* = 0.67$) to a maximum of 25 (for $D^* = 0.33$). These results, in absolute terms, in general are larger than those presented by Wright (2013), which referred to the displacement of the water within the tower as surge height. This could be linked to the difference in the initial water level in the shaft, which was significantly larger in the present study. An important result from this study is that even with $D^* = 0.67$

there were significant $\Delta Y_{FS}/D$ values, which is consistent with the observations by Wright (2013).

Fig. 2 presents the vertical trajectory of the air pocket interface and the free surface as they traveled up the vertical shaft. Representative test runs are presented for each combination of pocket volume and vertical shaft diameter. The errors related to both the time ($\approx 0.03 \text{ s}$) and the vertical coordinate measurements ($\approx 0.08 \text{ m}$) were small and are not represented in the figure. All cases showed a rise of both air–water interface and free surface, and the curved trajectories observed for $D^* = 0.33$ and $D^* = 0.50$ indicate that both free surface and interface velocities increased over time. However, for the single case in which geysers were not reported ($D^* = 0.67, V^* = 3.5$), the free surface level achieved a maximum elevation and receded prior to the breakthrough of the air pocket. This type of outcome has not been reported in earlier related investigations.

Fig. 3 presents pressure results measured from the moment when the air release occurred for a representative case involving a 202-mm-diameter shaft and $V^* = 7$, with pressure recordings

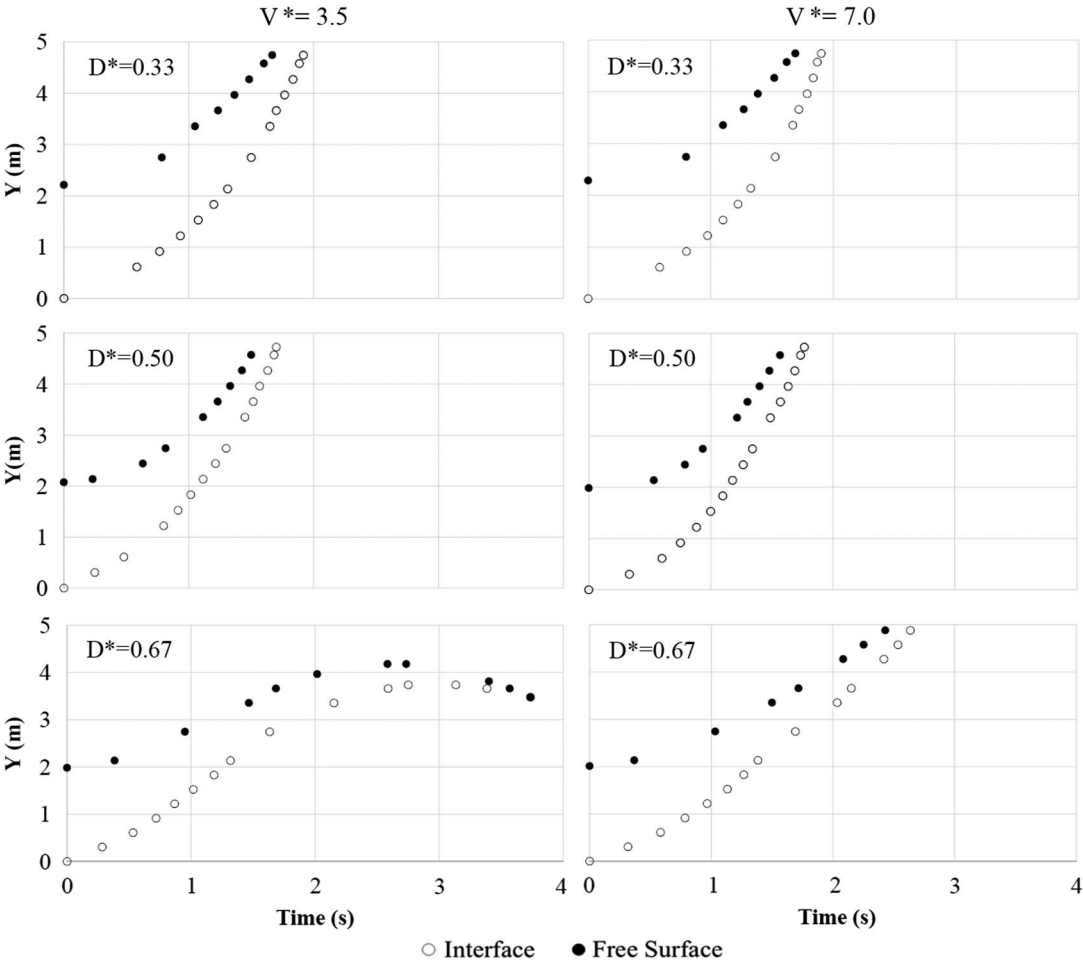


Fig. 2. Time evolution of free surface and interface coordinates for selected test runs for each tested condition

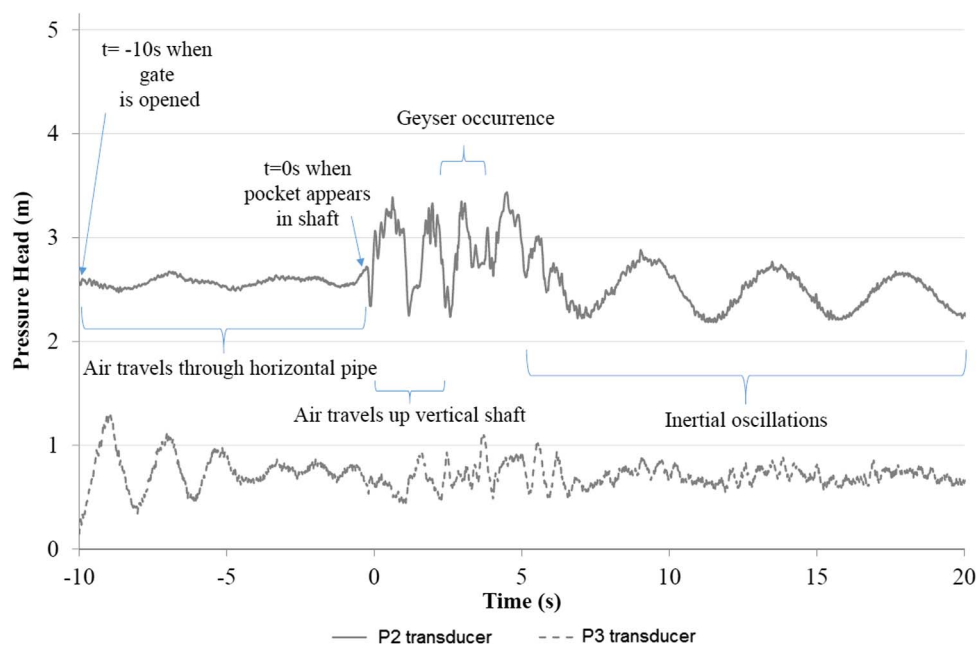


Fig. 3. Sample pressure head data from a 0.20-m-diameter shaft with a 200-L air pocket

at the bottom (Transducer P2) and side (Transducer P3) of the vertical shaft. The initial inertial oscillations were observed in Fig. 3 approximately 10 s prior to the instant when the air arrived in the vertical shaft. During the travel of the air pocket toward the shaft, pressure amplitudes in Transducer P2 were not as large as those recorded in Transducer P3. Smaller pressure amplitudes were expected in P2 because the point was closer to the reservoirs, which were sources of steady pressure. However, it also was speculated that the presence of the air pocket in the horizontal pipe, and thus closer to P2, could dampen the magnitude of the pressure oscillations generated by the water level change in the tower.

For that specific condition, this inertial oscillation period (approximately 3 s) was altered when the air reached the base of the vertical shaft, because pressure results presented more erratic patterns. Although the rise of the air pocket and free surface interface over time in general was monotonic, as indicated in Fig. 2, it is speculated that small changes in air–water interface velocities can lead to fluctuations in pressure. The pressure head recorded by P2 decreased during the air–water discharge (geysering) phase. This drop, however, was not as marked as those recorded by Transducer P3. After the geysering, air continued to migrate upward in the shaft and the complex bubbly flow eventually approached single-phase conditions, with more-regular mass oscillations at the end of the pressure hydrograph reflecting this final condition as the reservoirs refilled the vertical shaft. Finally, the amplitude of the pressure fluctuations recorded in P3 was not as large as those in P2, and neither of the piezometric pressure head values were large enough to reach the elevation of the vertical shaft rim.

Fig. 4 presents the normalized velocity results for the air–water interface and free surface calculated from the measured displacement of the flow features. These results indicate a slight decrease between $D^* = 0.33$ and $D^* = 0.50$, and a more pronounced velocity drop for $D^* = 0.67$. The velocity for $D^* = 0.67$ and $V^* = 3.5$ became slightly negative, consistent with the receding free surface interface observed for this condition. In absolute terms, the velocities of rising air pockets were slightly higher for the $D^* = 0.50$ condition than for $D^* = 0.33$. The velocity results for $D^* \leq 0.50$ in general were characterized by steady growth, but this was

influenced by a *wobbling* that may be linked to compression and decompression of the air pocket as it moved upward. Additionally, conditions resulting in geysering presented much faster growth of the air–water interface velocity as soon as geysering began. It is assumed that once the water is discharged at the top of the vertical shaft there is a rapid increase in the pressure gradient experienced in the air phase, leading to even faster upward motion.

Fig. 5 shows a set of representative pressure data for the three diameters and the two air pocket volumes studied measured by Transducer P2. The data sets were trimmed to display only the data for the period during the upward water motion in the vertical shaft. The range of pressure fluctuations decreased with the shaft diameter D ; however, in no condition did the pressure head exceed levels comparable with the vertical shaft rim. In the cases in which geysers were observed, they occurred between 2 and 3 s, but no clear trend in the pressure values was identified during these releases.

Fig. 6 presents a sequence of photos of the progression of a typical geyser event with relative time stamps, in which T_{rel} is measured from the time when the air reached the visible part of the vertical shaft ($Y_{int}^* = 0$). In Fig. 6(a), the free surface was approximately 0.1 m from the rim of the shaft ($Y^* = 0.95$) and the leading edge of the air pocket was approximately 1 m below at $Y^* = 0.80$. Fig. 6(b) corresponds to the instant at which the free surface was at the rim of the shaft, $Y_{FS}^* = 1$ and the air pocket was 0.75 m below ($Y_{int}^* = 0.85$). This frame marks the onset of a geysering event. As shown in Fig. 6(c), the geysering event progressed as water above the air pocket continued to be expelled like a slug until the air pocket broke through to the atmosphere ($Y_{int}^* = 1$) approximately 0.1 s later, as captured in Fig. 6(d). Although there was no measurement of air velocity in these discharges, the CFD-simulated air pocket velocity in Fig. 6(d) was 8.15 m/s, and in Fig. 6(e) it was 18.36 m/s. Figs. 6(e and f) show a very complex interaction between the discharged water and the air that is being discharged, which included a fragmentation of the water slug that spread the air pocket several meters in the air. Such air velocities are feasible due to a short-duration, albeit strong, pressure gradient in the shaft that is speculated to occur in the shaft as soon as air pocket

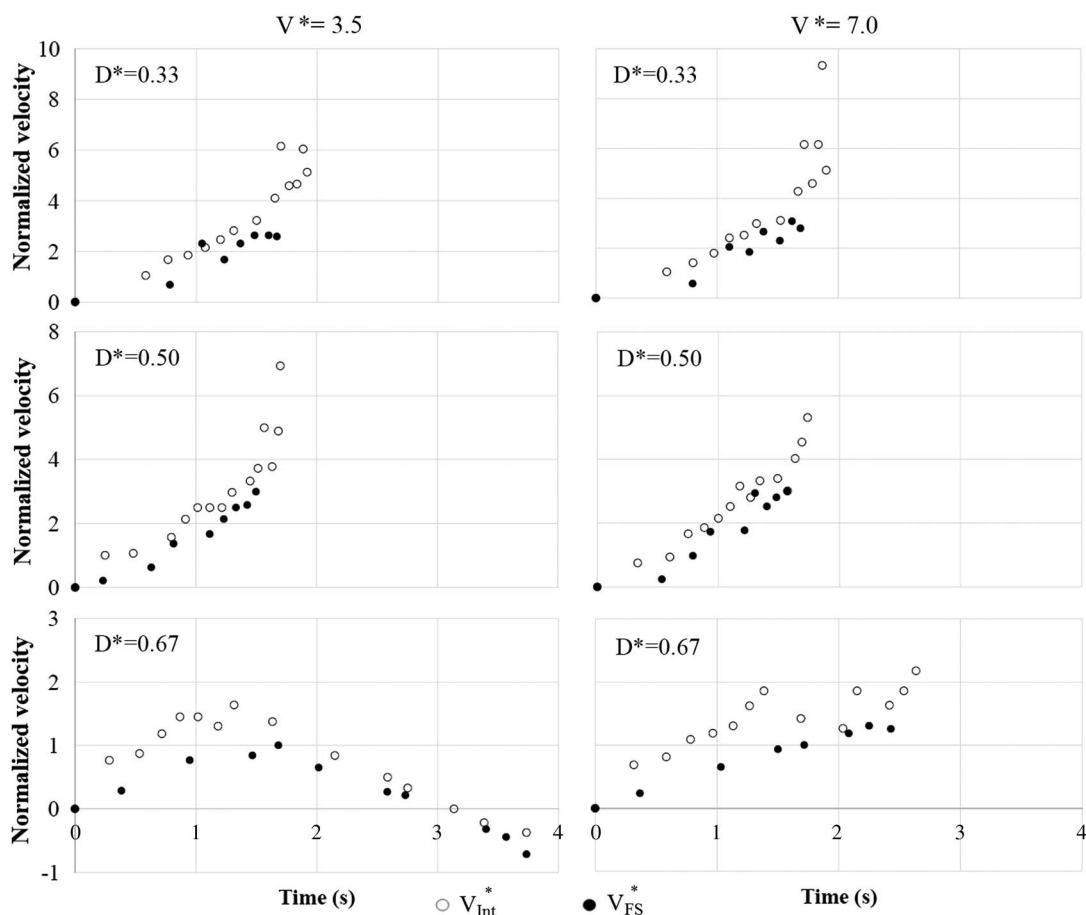


Fig. 4. Time evolution of free surface and interface normalized velocities for selected test runs for each tested condition

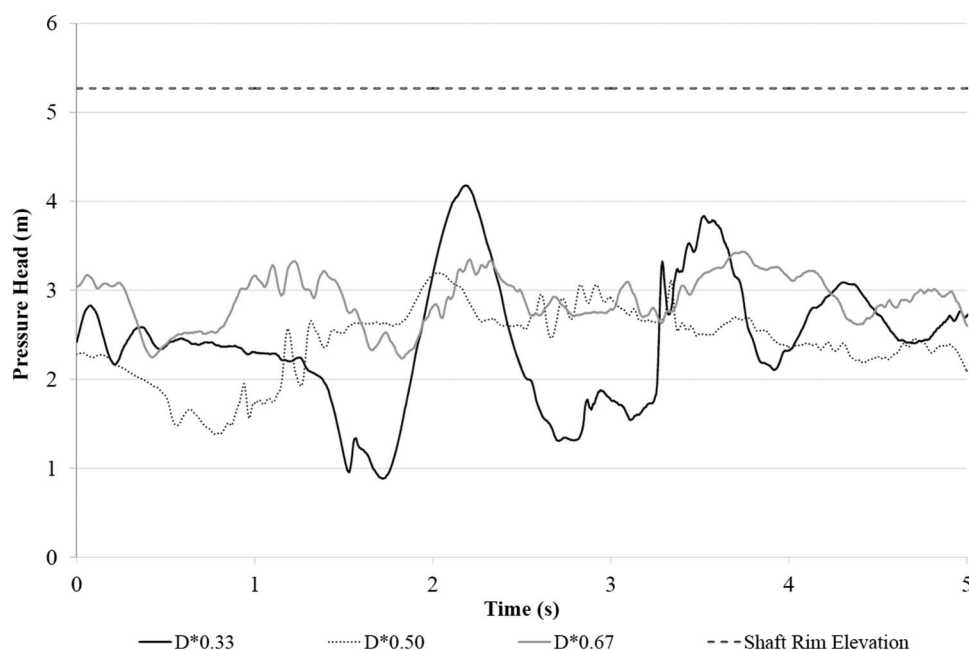


Fig. 5. Pressure head results from Transducer P2 (bottom of shaft) from three shaft diameters with air-pocket volume $V^* = 3.5$

breakthrough occurs. Although it was not easily observed, a type of *mist* following the water slug discharge was observed in different tests. It is assumed that flooding instability, such as presented in Guedes de Carvalho et al. (2000), potentially could have occurred

between the film flow moving downward along the pipe walls and the air pocket moving upward. As water moved downward during the air release, a fraction of it could have detached and entrained with the upward-moving air pocket.

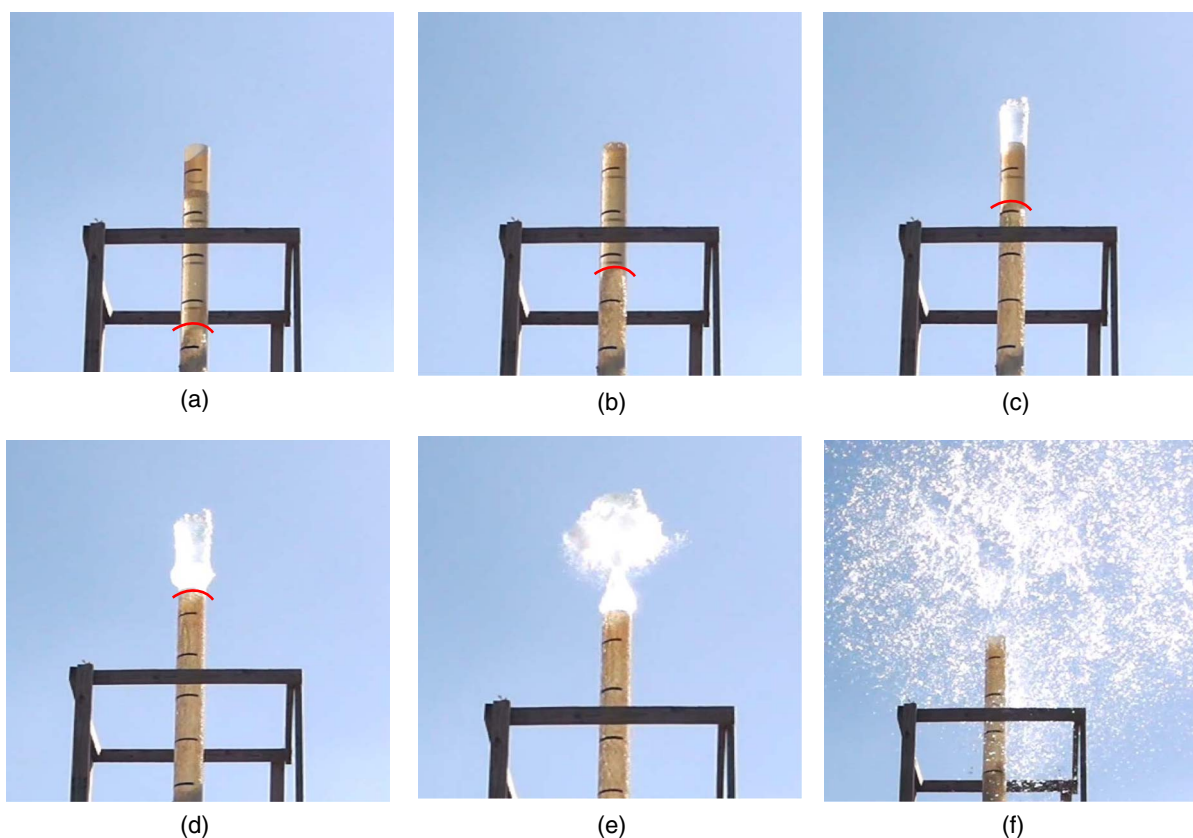


Fig. 6. Progression of a geysering event ($D^* = 0.5$, $V^* = 7.0$) from the time the pocket is visible; the air–water interface locations are highlighted with curved lines; markers in the shaft are 0.3 m apart; simulated air velocity as it exited the shaft exceeded 18 m/s, explaining the explosive nature of this event

CFD Modeling Results

Experimental Conditions Modeling

The initial work involving CFD aimed to assess the ability of the *compressibleInterFOAM* solver to represent the experimental conditions. A CFD model was applied for all experimental cases, including three vertical shaft diameters and both 100-L and 200-L air pockets. This paper selected the condition with $D^* = 0.50$ (150 mm) and $V^* = 3.5$ (100 L) to illustrate the characteristics of the numerical solution. In general, the other simulated conditions yielded comparatively good agreement with the experimental data, particularly with respect to the simulated displacement of the rising water free surface and air–water interface.

Fig. 7 presents the comparison of the normalized displacement of the water free surface in the shaft (Y_{FS}) and air–water interface of the rising air pocket (Y_{Int}). Time is referenced to the instant when the air pocket leading edge became visible in the vertical shaft. The CFD modeling results matched the measured data well, considering the natural variability among experimental repetitions. This agreement improved with the discretization, as seen when comparing results obtained with the coarse mesh with intermediate and fine mesh results. For both the CFD and experimental results, the air pocket and water column above the pocket rising process ended within 2 s, except for the results obtained with the coarse mesh.

Fig. 8 compares pressure head measured by Transducer P3 (normalized by the shaft length L) with CFD predictions at that location. In this run, water levels at the shaft were 1.0 m above Transducer P3 as the air pocket arrived in the shaft. Simulated pressures presented the general trends observed during the experiments,

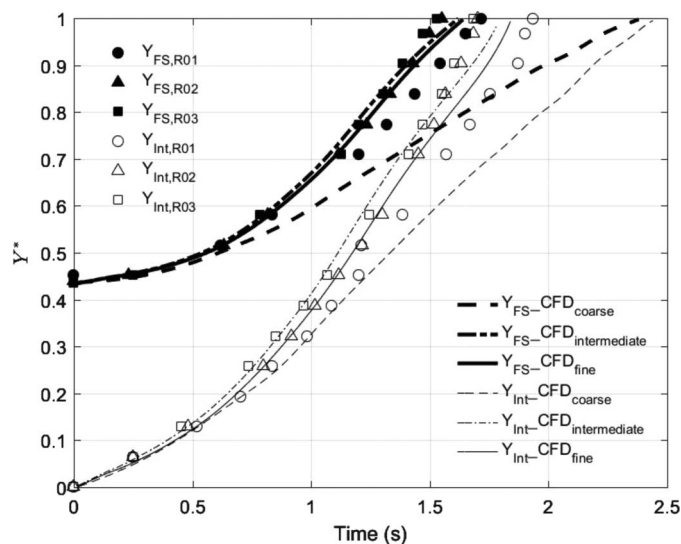


Fig. 7. Normalized free surface and air–water interface displacement

including the sharp pressure drop and rise after the pocket arrival at the shaft and the trend of maximum and minimum pressure head oscillations. Results with the intermediate and fine meshes were similar, and in general yielded a closer match to the events observed in the experiments than did the coarse mesh results. Furthermore, CFD-simulated pressure head results never exceeded the shaft rim elevation, even during geysering occurrence. The CFD pressure results were not as accurate as the simulation of the Y_{FS} and Y_{Int} ,

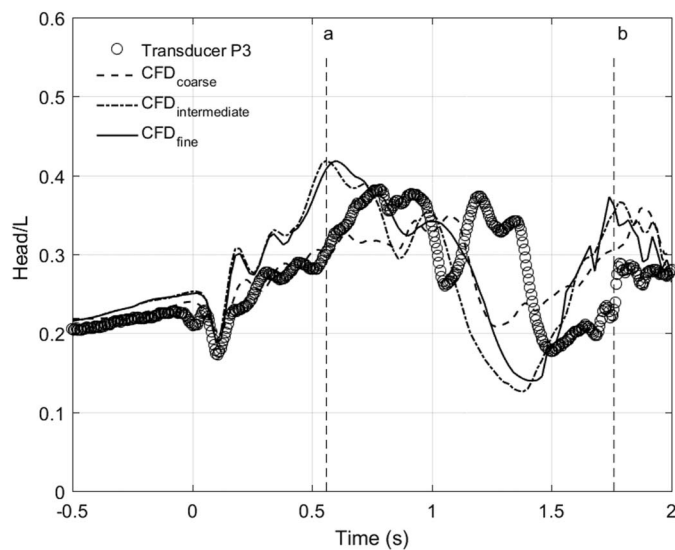


Fig. 8. Normalized pressure head at Transducer P3 on the shaft side; vertical dashed lines refer to the time in the simulations when (a) air pocket leading edge reaches Transducer P3 elevation; (b) air pocket breakthrough at shaft rim

because CFD overpredicted the magnitude of pressure oscillations. There also was a time offset between the CFD prediction of the air pocket arrival at Transducer P3 and the air pocket breakthrough at the shaft rim ranging from 0.1 to 0.2 s.

Results from this evaluation were considered positive, particularly with regard to the predictions of the free surface and interface coordinates during air pocket release. This led to the second stage of numerical simulations, involving air releases in large geometries.

Simulation of Air Pocket Release in Large Geometries

As stated in the Methodology section, this paper used CFD to perform a systematic evaluation of the effects of air pocket releases involving geometries that were larger than those used in the experiment. The geometry used in this stage of the CFD result was in

general similar to the one used in experiments, except that the upstream reservoir was absent (Fig. 9). The reservoir upstream was removed simply to reduce the size of the computational domain and thus speed up computation. Air pockets would be released from this upstream end, at the opposite end from the water reservoir that sustained the pressure in the system. This study considered varying shaft diameters, initial water levels, and air pocket volumes using the CFD model developed in the stage of experimental results comparison. Table 1 presents the range of tested conditions. The mesh size: tunnel/shaft diameter ratios used in this phase of the work were the same as those used in the intermediate mesh resolution of the CFD simulations representing the experimental conditions.

The choice of air pocket volumes to be included in the numerical simulations was based on Vasconcelos and Wright (2016). Air pockets volumes were categorized as small ($V^* < 2$), medium ($2 < V^* < 10$), large ($10 < V^* < 50$), and very large pockets ($V^* > 50$). As shown in Table 1, this stage of CFD simulation considered four pocket volumes ranging from small to very large air pockets. Simulation results involving $V^* = 1$ did not indicate any significant displacement of Y_{FS} in shafts for all D^* . Such results therefore are not included in the present discussion. In addition, simulations with $V^* = 50$ were performed only for three cases: (1) $D^* = 0.50$, $Y_{FS,0}^* = 0.067$; (2) $D^* = 1.0$, $Y_{FS,0}^* = 0.133$; and (3) $D^* = 1.00$, $Y_{FS,0}^* = 0.333$. These tests with very large pockets needed a longer water-filled horizontal tunnel length of 110 m, with the location at which the pocket was released 80 m from the base of the vertical shaft.

Fig. 10 presents some typical flow patterns at the shaft junction as the air pocket enters the shaft. The three conditions represented are (1) $D^* = 0.25$, $Y_{FS,0}^* = 0.133$, and $V^* = 10$; (2) $D^* = 0.50$, $Y_{FS,0}^* = 0.133$, and $V^* = 10$; and (3) $D^* = 1.00$, $Y_{FS,0}^* = 0.133$, and $V^* = 10$. The small shaft diameter ($D^* = 0.25$) did not capture larger fractions of the air pocket traveling in the horizontal tunnel, and the captured fraction increased with the shaft diameter. Fig. 10 does not show the maximum thickness of the rising air pocket core because it occurred along the centerline of the shaft.

The outcome is reflected in the amount of air mass in the vertical shaft when breakthrough occurs at the shaft rim, shown in Fig. 11. Results in this figure are normalized by the initial amount of air in the system in the beginning of the simulation. Depending on the

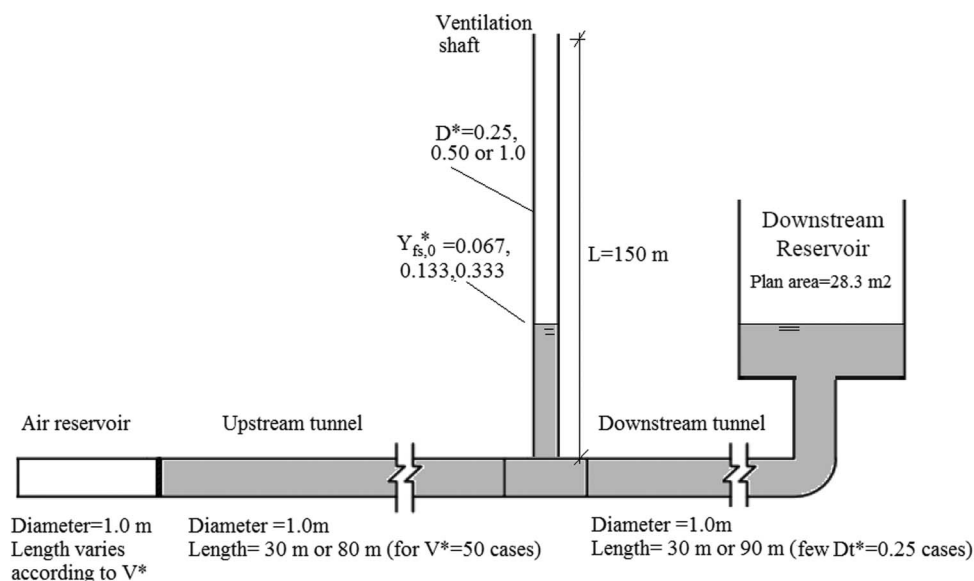


Fig. 9. Geometric characteristics of the large-scale CFD model; conditions in the figure are representative of the start of the simulation

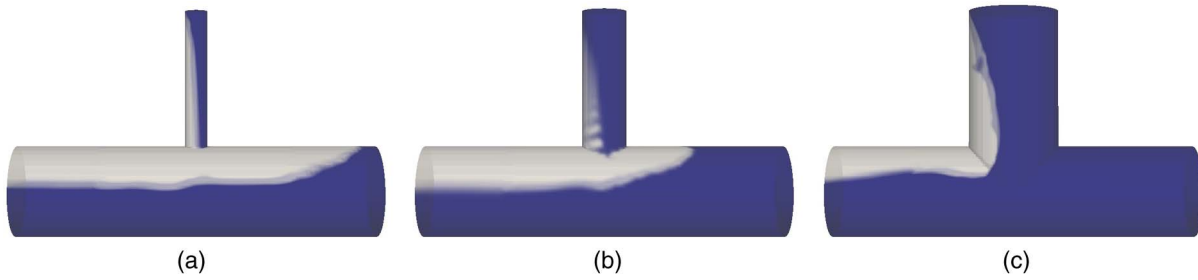


Fig. 10. Typical early-stage air release patterns at shaft junction for large-scale simulation predicted by *compressibleInterFOAM*

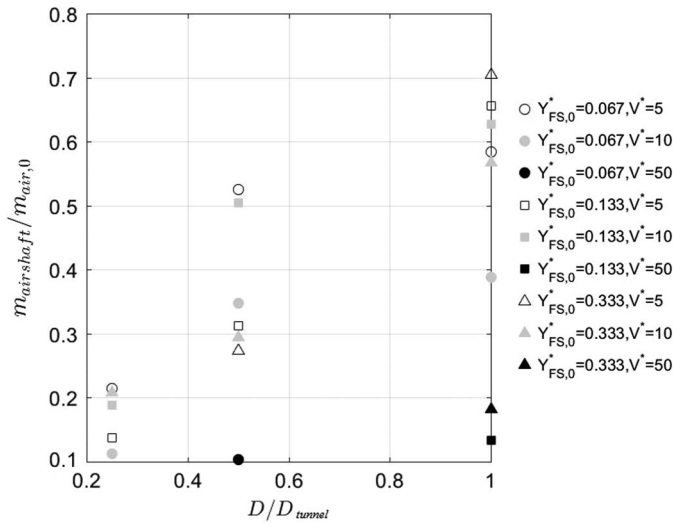


Fig. 11. Normalized mass of air pocket captured in the shaft prior to air pocket breakthrough

initial volume of the air pocket and the initial water level in the shaft, the air mass fraction in the shaft ranged from 0.11 to 0.22 for $D^* = 0.25$, from 0.10 to 0.53 for $D^* = 0.50$ and from 0.14 to 0.71 for $D^* = 1.0$. There are three important remarks related to results presented in Fig. 11. First, lower air mass fractions in the shafts may be linked in real stormwater systems to occurrences of multiple geysers, considering that the simulations performed here were interrupted soon after breakthrough occurred. Second, small fractions of air mass in the shaft were noticed when very large air pockets ($V^* = 50$) were released, which again may be an indication that some large pockets may be ejected through a sequence of geysering events occurring in succession rather than a single large geyser event. Finally, for given values of V^* and $Y_{FS,0}^*$, the fraction of air within a shaft increased with D^* , a result that is intuitive based on the flow patterns presented in Fig. 10.

One key objective of this work was to estimate the amount of water displacement during uncontrolled air release in shafts. Fig. 12 presents simulated results of maximum incremental rise of the free surface ΔY_{FS} , normalized by $Y_{FS,0}$. Values of $\Delta Y_{FS}/Y_{FS,0}$ were influenced by the shaft diameter D^* , a result that is consistent with the experimental tests and with previous investigations. Although $\Delta Y_{FS}/Y_{FS,0}$ calculated with D^* values of 0.25 and 0.50 were similar, this value decreased significantly for $D^* = 1.0$. These simulated $\Delta Y_{FS}/Y_{FS,0}$ potentially can be useful in estimating the maximum rise of water rise in stormwater shafts during uncontrolled air releases and assessing whether these displacements have the potential of raising the free surface to grade and triggering geysers.

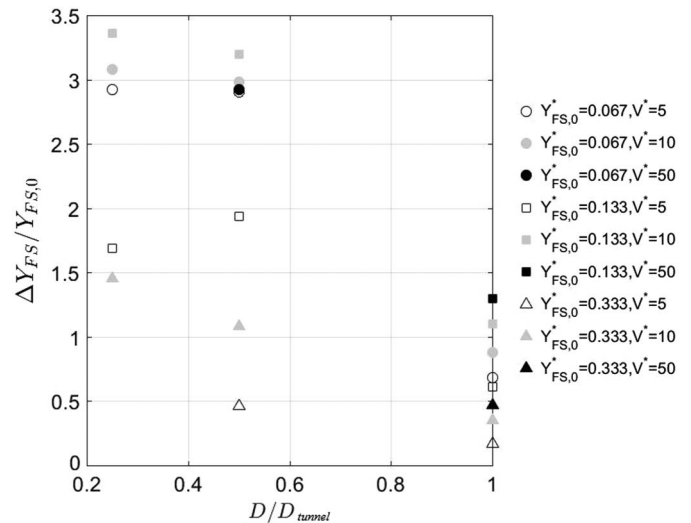


Fig. 12. Normalized maximum incremental free surface displacement

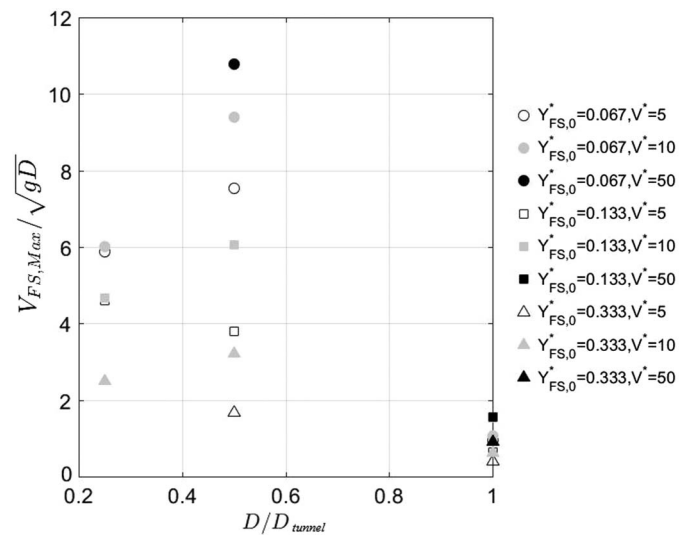


Fig. 13. Normalized maximum free surface velocity

Fig. 13 presents the maximum upward free surface velocities in the vertical shafts. These values, which are normalized by $\sqrt{gD}$, are useful because they are related to the air displacement within shafts and potentially to manhole lid displacement. Results presented in Fig. 13 indicate that highest maximum velocities were observed for smaller $Y_{FS,0}$, a result that was expected because lower initial water levels are linked to smaller inertia and would be more easily

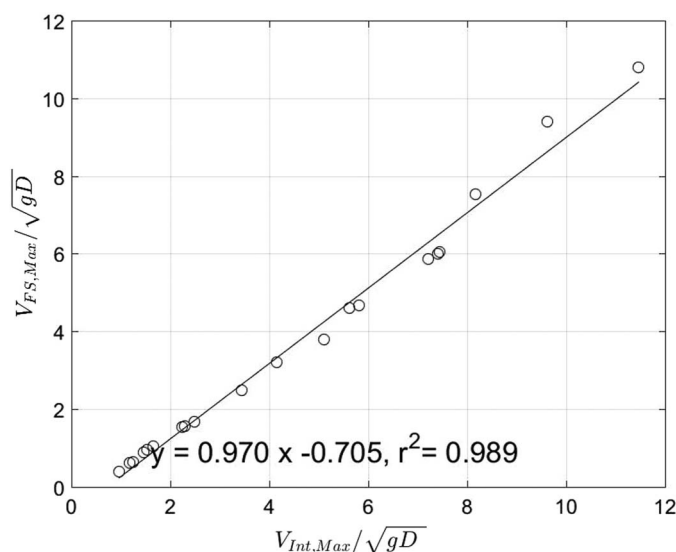


Fig. 14. Normalized water–air interface velocity versus free surface velocity

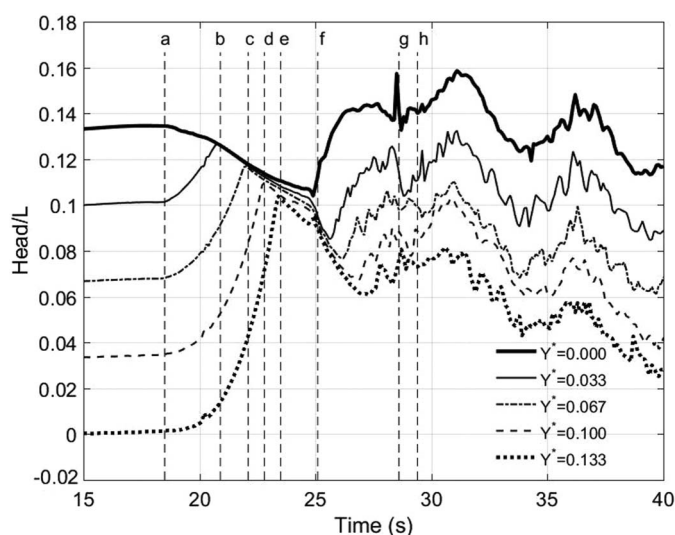


Fig. 15. Normalized pressure head at different shaft elevations for case $D^* = 0.50$, $Y_{FS,0}^* = 0.133$, $V^* = 10$

accelerated. The maximum free surface velocity increased with air pocket volumes, and it is speculated that this could be linked to the effect of decompression of larger air pocket volumes to the free surface location. Finally, an interesting result was that the highest normalized free surface velocities were observed for $D^* = 0.5$ and $Y_{FS,0} = 0.067$, with maximum velocity above 24 m/s within the shaft. It is possible that such an outcome is linked to the ability of this shaft diameter to discharge larger fractions of air. There was a good agreement between the maximum velocity of the free surface calculated with the CFD model and the corresponding interface velocity $V_{Int,Max}$. The best-fit equation ($R^2 = 0.989$) relating $V_{Int,Max}$ and $V_{FS,Max}$, presented in Fig. 14, is

$$V_{FS,Max} = 0.970V_{Int,Max} - 0.705 \quad (6)$$

Numerical simulations also were used to study the change in pressure at different locations in the vertical shaft as air pockets

were admitted then expelled. Fig. 15 presents the results for the case when $D^* = 0.50$, $Y_{FS,0}^* = 0.133$, $V^* = 10$, in which L is the physical length of the shaft. The following events were identified in the figure during the air release: (a) air pocket enters the shaft; air pocket leading edge reaches elevations; (b) $Y^* = 0.033$; (c) $Y^* = 0.067$; (d) $Y^* = 0.100$; (e) $Y^* = 0.133$; (f) water starts to refill the base of the shaft; (g) water free surface reaches the maximum elevation; and (h) air pocket breakthrough occurs. In this simulation, the maximum water level reached by the free surface was 83.8 m.

Pressure results indicate that before 18 s the pressure levels were steady in the shaft, and the varying pressures reflected the quasi-hydrostatic conditions. In the early stages of the pocket advance, between 18 and 25 s into the simulation, the air pocket delimited two regions in the flow. From the base of the shaft to the leading edge of the air pocket there was the air phase, which showed a decrease in the pressure head of approximately 0.03 L. Prior to the arrival of the air pocket to the elevation, simulated pressures increased because of the displacement of the water surface ahead of the pocket motion. Between Markers e and f a difference in the air phase pressure head of 0.01 L was noticed.

At approximately $T = 25$ s, the tail of the horizontal air pocket reached the base of the shaft, and water started refilling the shaft. A complex flow pattern was observed at lower elevations of the shaft, with a mix of air and water still rising in the flow. Pressure recovery was detected first at the base of the shaft, and this recovery propagated over a short time to the locations at higher elevations. This complex flow pattern continued from this point onward. After 30 s in the simulation there was no identifiable large air pocket in the shaft, but only a complex air–water bubbly mix that underwent inertial oscillations while discrete bubbles migrated upward.

Conclusion

This paper presented results from large-scale experimental tests and CFD simulation of the release of large air pocket volumes through water-filled shafts, triggering geysering episodes. Although such occurrences are severe and cause serious concern in actual stormwater systems, much remains unknown about geysers, and research efforts such as this still are warranted. Compared with most studies presented to date, this paper used larger diameters for the horizontal and vertical components of the apparatus. The release of air enabled the observation, in some cases, of strong discharges that exceeded several meters in height and had an explosive nature. In other cases, a mist of very fine water particles was observed when the water slug discharge was complete. These characteristics were not reported in previous studies performed at smaller scale, such as Vasconcelos and Wright (2011).

The understanding of the kinematics of the free surface motion within the shaft and the air–water interface motion is of paramount importance to designers of stormwater systems concerned with the possibility of geysers. The experimental tests indicated significant displacements of water in junctions, even for the case in which $D^* = 0.67$, which is consistent with Wright's (2013) studies. However, the displacements reported here were larger than those measured by Wright (2013), possibly because of the larger initial water level in the vertical shaft. A better understanding of the water displacement created by air pockets of various sizes and rising in varying shaft geometries will lead to improved safety in stormwater system designs.

Pressure head measurements during experiments indicated that levels were always below the shaft rim during geyser events,

which is consistent with previous investigations. Drops in the pressure measured by Transducer P2 at the bottom of the shaft were observed during geyser occurrences. These drops were not as identifiable through the pressure measurements at the lateral of the shaft by Transducer P3. This result confirms earlier experimental studies that showed that geysering events can occur when pressure heads are well below grade elevation.

A CFD model using *compressibleInterFOAM* solver was created to represent the same geometry as in the experimental conditions. The tests provided good agreement regarding the kinematics of the water and air–water interface during the air pocket release. The comparison between measured and simulated pressures was not as accurate, although the general trends were well represented. The CFD model then was used to simulate the release of air pockets in various conditions involving much larger diameters. The idea was to focus on the potential displacement created by the air pocket release, and therefore the vertical shaft was made long. The model indicated that the fraction of the original air pocket in the system that is released increased with the shaft diameter. The displacement of the free surface also varied with D^* , with significant $\Delta Y_{FS}/Y_{FS,0}$ values for $D^* = 0.25$ and $D^* = 0.50$ but with decreased values for smaller air pocket volumes. Finally, the velocity of free surface motion increased with air pocket volume and smaller initial water levels in the shaft. These kinematic results of free surface maximum rise and velocity are potentially useful in the context of design of stormwater systems to avoid geysering episodes.

There are significant knowledge gaps yet to be addressed in future research. For instance, it would be relevant to measure the pressure within the vertical shaft (i.e., centerline) during the air release phase and compare results with CFD results. Other modeling approaches should simulate the air release applying CFD, for instance adopting more sophisticated turbulence modeling approaches, and more systematic study of the effects of wall roughness in the simulation results also is needed. Other interesting large-scale conditions still to be simulated with CFD should include situations in which there is an *air-core* condition, the case described by Wright et al. (2011) in which the air pocket break breaks through the upper surface but still is present in the base of the shaft. Large pressure gradients should be observed in such cases, and the intensity of geysers represented in the CFD simulations are expected to be impacted. The use of large-scale tests to study the nature of air release in water-filled shafts, both in laboratory and in numerical simulations, are anticipated to have greater priority in future research efforts, along with studies to mitigate geysers in actual stormwater systems.

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Notation

The following symbols are used in this paper:

- C_p = specific heat;
- D = vertical shaft diameter;
- D_t = horizontal tunnel diameter;
- F_S = surface tension force;
- g = gravity acceleration;

- k = thermal conductivity;
- L = length of vertical shaft;
- m_{air} = mass of air phase;
- p = pressure;
- S_T = energy source;
- T = temperature;
- t = time;
- $\mathbf{U}$ = velocity vector;
- U_r = velocity field suitable to compress the interface;
- V = air pocket volume;
- V_{FS} = velocity of the water free surface within shaft;
- V_{Int} = velocity of the air–water interface;
- Y = vertical coordinates;
- Y_{FS} = vertical coordinate of water free surface within shaft;
- $Y_{FS,0}$ = initial coordinate of water free surface within shaft;
- Y_{Int} = vertical coordinate of air–water interface;
- α = volume fraction of water;
- ρ = average density of air–water mix;
- ρ_a = density of air phase;
- ρ_w = density of water phase; and
- μ = dynamic viscosity.

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